

Hybrid deep learning and machine learning approach for passive image forensic

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Abhishek Thakur¹ ✉, Neeru Jindal¹

¹Electronics and Communication Engineering Department, Thapar Institute of Engineering and Technology, Patiala, India

✉ E-mail: abhishek@thapar.edu

Abstract: Image forgery detection using traditional algorithms takes much time to find forgeries. The new emerging methods for the detection of image forgery use a deep neural network algorithm. A hybrid deep learning (DL) and machine learning-based approach is used in this study for passive image forgery detection. A DL algorithm classifies images into the forged and not forged categories, whereas colour illumination localises forgery. The simulated results are compared to other algorithms on public datasets. The simulated results achieved 99% accuracy for CASIA1.0, 98% accuracy for CASIA2.0, 98% accuracy for BSDS300, 97% accuracy for DVMM, and 99% accuracy for CMFD image manipulation dataset.

1 Introduction

The means of socialism and connectedness, nowadays, is achieved by uploading their images on social media. The availability of personal pictures on social media has also made readily available for other people, and they can manipulate these private photos. So, the concept image forensics come into the picture to check the authenticity of these images. There is a great importance of photos in the courtrooms and journalism. Many researchers have been continuously working within the area of image forensic. Photographs and other electronic media have shown their importance in many legal cases. These images and other electronic media considered as proof against someone in the legal cases required certification of originality, faith, and truth. Offence moments caught in the camera needs to verify its originality and truthfulness in a courtroom of law. Hence, in that respect is a unique need for preserving digital media. The software tools have modified these images, an easy task. One does not even need to have the complete knowledge of employing software to add tempered regions to alter the images. The most common type of forgery that is performed in every corner of the world is retouching, copy-move, and splicing the image effects. Other vital applications where the processing of images is done include object recognition, segmentation [1], video forgery detection, and image forgery detection. The primary divisions of forgery detection methods are active and passive detection methods. The one where the source image information is needed for the further processing of the image is called the active technique. Its example can be taken as digital watermarking and digital signatures. Whereas the latter method does not accept any prior information for the processing on the image is called a passive technique. The traditional methods for image forensics like block-based or key-

point are time consuming. Hence we require to come up with some other idea, which is less computationally complex and requires no continuous human interaction. Automated machine learning (ML) gave hope to achieve these aims. However, the traditional training algorithms for forensics using ML were highly sophisticated. In an algorithm primarily based on machine getting to know and profoundly gaining knowledge along to discover copy move and splicing forgeries (SFs) [2].

The predominant targets of this work are (i) to design a deep neural network (DCN) using supervised learning, (ii) to classify and find out forgeries for large datasets, i.e. SF or copy-move forgery (CMF), and (iii) use an ML knowledge of the algorithm to realise forgery.

1.1 ML technique

ML recognises an object based on features. If labelled data is given to training the network, it comes under supervised learning. If labelled data is not given to training the network, it comes under unsupervised learning. Firstly, the forged features are pulled out from the labelled dataset. Then these extracted features are applied for the learning algorithm to train the network. At the test phase, the classifier predicts output by comparing the features of test images and learned image features. In supervised learning, the labelled dataset is available for all images. A support vector machine and linear component analysis are used to perform this task. These classifiers learn with the help of labelled data in which an image is forged or original.

As we can see in Fig. 1, training images are taken to the input block. The features of these images are extracted. These features are applied to machine learning algorithms along with the label. In this machine learning algorithm, the human need to extract features. During the test phase, the same operation is performed, and the predictive model gives the output as expected level.

1.2 Deep learning (DL) technique

The first DCNN layer incorporates the pixel values of the images. Normalisation is achieved by subtracting the second image of the training set from the input image. In the convolutional layer, 32 filters are used. Consider the small patch of image and use dot product by sliding over the image. A series of cognitive operations are applied to every layer. If these layers are more than one, then the network is called DNN, as shown in Fig. 2.

The objectives of this paper are

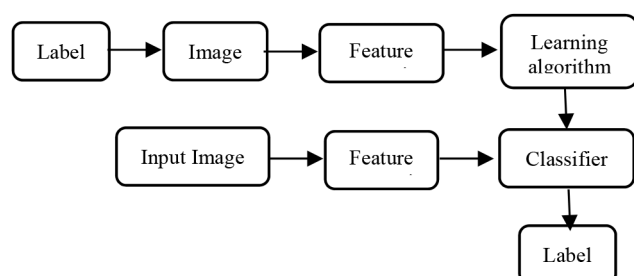


Fig. 1 Block diagram of a ML-based algorithm

- (i) To design a DCNN for classification between forged and not forged images from a large dataset.
- (ii) To implement an ML-based colour illumination approach to detect forgery.
- (iii) Hybrid DL and ML-based models are trained using supervised learning and tested on different datasets.

In Section 2, the literature survey is conducted on forgery detection in copy-move and splicing using machine learning and DL CNN. In Section 3, the flowchart and block diagram of the proposed algorithm are discussed in detail. In Section 4, results have been shown by using different public benchmark datasets such as CMF and splicing images [3, 4]. In Section 5, the conclusion is given.

2 Literature survey

Bunk *et al.* [5] proposed two different forgery detection techniques. Firstly, they utilised resampling features and DNNs to detect the tapered region. They utilised random walker segmentation for the localisation of the forgery. Secondly, those resembling features are further used for classification when passed through a long short-term memory (LSTM). Tarman [6] introduced a technique for the detection of CMF in the mirror rotated images based on the improved scale-invariant feature transform (SIFT) method, i.e. M-SIFT. The regions are localised with the accuracy of 98%, but the time is taken for computing this tempering algorithm is quite large. Fengli and Qinghua [7] introduced a method to detect the false generation of power frequencies in grids using neural networks. The Fourier transforms used to detect forgery attacks in automatic generation control (AGC). They used area control error (ACE) time series patterns to find irregular patterns caused due to forgery. Thirunavukkarasu and Kumar [8] proposed a passive copy-move forged region detection using fast retina keypoint descriptor (FREAK). FREAK descriptors are extracted across the Harris corners that are then mapped to the K -means algorithm. Cheng and Meng [9] gave the method for optical remote sensing harbour images. A convolutional network is applied to observe the objects like sea, ground, and ship by learning the edge-features. The training of the network is performed by using edge detection networks. Firstly, different extracted features from the defined segmentation network are used to train the edge network. Secondly, the outputs of the edge trained networks are further fed to improve the whole model. This process is carried out by the addition of an edge-aware regularisation to the accurate, efficient, and consistent results when coming to the shape formation. Li and Liu [10] showed in their algorithm that SIFT features could be effectively used to minimise the distortion in the image that has been processed a multiple number of times. While removing the SIFT key points, the traces are usually left behind, and these are removed by injecting a large number of fake SIFT key points in the last generated image to minimise the distortion. The results are better than other methods. Husain *et al.* [11] proposed a technique to detect the scene objects clearly by using DNNs. This technique is based on learning the pre-trained geometrical feature's pixel-wise and labels the specific objects to detect the indoor scenes precisely. Amerini *et al.* [12] proposed a SIFT-based keypoint detection method that localises the copy-move part by using the concept of geometrical transformations. Li *et al.* [13] gave the technique for hyperspectral image classification based on the concept of DL. They showed how working with large databases is a problem when we use traditional methods. The proposed work is carried out on the uncertainty sampling strategy. Stacked autoencoders are trained on spatial and spectral features. These features are used to train the stacked autoencoders and initialise a DNN. Nithiya and Veluchamy [14] extracted the same kind of forgery by using the adaptive over-segmentation. The high computational complexity could have been reduced by creating non-overlapping blocks of the image. Zhao *et al.* [2] showed that the complex images and the irregular patterns in the image make a complicated task to detect the forgery precisely when worked with the traditional methods. However, DL has reduced the complexity way far. Furthermore, to reduce the problem of detecting the forgery at the pixel level with neural networks, they proposed an object-based DL method for the

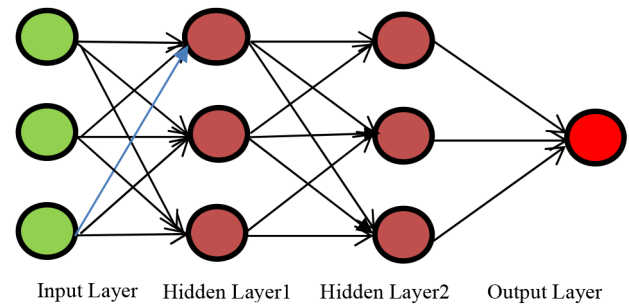


Fig. 2 DCNN operation

precise classification of the images with the high-resolution even with less human involvement. They extracted the high-level feature through the network over five different defined layer configurations. Wang and Shen [15] gave a DL model for the detection of salient regions within the frames of videos. The so proposed saliency network based on DNN data training consists of two segments, each one for detecting the spatial and temporal saliency materials, respectively. The time for detecting the saliency features with neural networks is way faster than the traditional optical flow method. To prevent overfitting, the augmentation technique with a limited number of training videos were given that improves the accuracy further. Pagnutti *et al.* [16] introduced a technique for the segmentation and semantic labels of RGBD data. After the first step of over-segmentation using spectral clustering, the extracted blocks fed to the nine convolutional layered neural networks that contain the colour and geometry data. The so defined softmax produces a vector of descriptors for each sample. Another algorithm defined in the next step recombines the over segmented output. The segments with higher, CNN feature similarities are used to find out the regions that belong to the same surface. Chen *et al.* [17] gave the idea of using convolutional neural networks (CNNs) to detect the median filtering from small-size of the compressed regions of the image by automatically learning and obtaining the features directly from the given image. Girshick *et al.* [18] proposed a region-based convolutional network detection algorithm to improve mean average precision by introducing two different ideas. In the foremost one, high-capacity convolutional networks (CNNs) are employed to get the segment objects, and in the second, supervised pre-training, the network is made during the scarcity of labelled training data. Zhan *et al.* [19] introduced an idea for learning the super pixel-based change in high-resolution images using neural networks. Superpixel segmentation generates highly compact superpixels to find the image boundaries easily. The changed features represent any conflict in the information about the texture in the icon. The DNN is then employed to determine the semantic difference between the pixels that were altered and those which were not. Keglevic and Sablatnig [20] arrived at an algorithm for assessing the similarities between striated tool marks. They studied the individual characteristics of the different forensic tools by varying parameters. Automated CNN was built on knowing any similarity in striated tool marks, and these are mapped accordingly. Chen and McCloskey [21] used the camera response functions (CRFs) to detect the copy-move part in the image. The analysis of the non-linear CRFs showed that intensity gradient bivariate histograms of the edges are affected by the CRFs. A deep-learning method is defined to localise forged edges. Gao *et al.* [22] proposed a technique by using ConvNets for visual data recognition tasks. The learning of neural networks is managed along with the basis of the large labelled training set. The label of imagery present in the data sets is changed to a discrete label distribution, which is further determined by the minimisation of the Kullback–Leibler divergence by knowing the difference in the values of the ground truth images and the ConvNet predicted. Classifier learning is used for the prevention of overfitting during the small training sets. Rao and Ni [23] proposed a technique to learn colour features automatic on the DCNN. These features are learned by initialising the neural network with the appropriate weights and high-pass filter. The CNN model is furthermore utilised to extract solid features. Also, the feature fusion technique

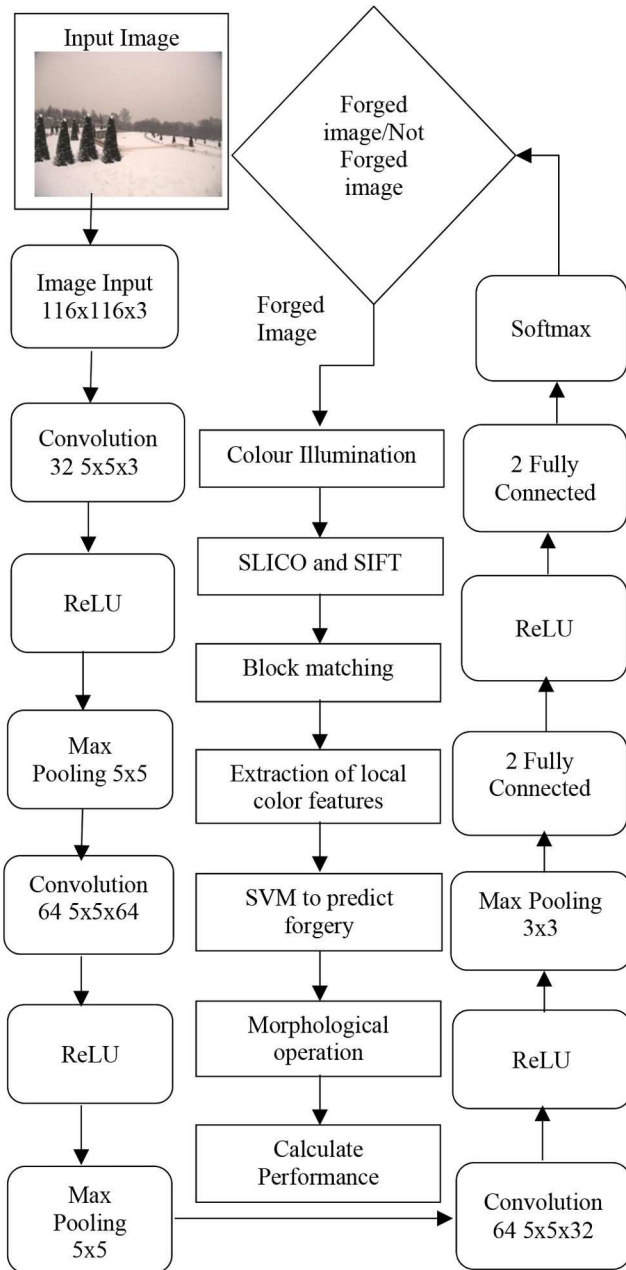


Fig. 3 Block diagram of hybrid DL and ML approach for passive image forensic

is used to classify the images with the SVM classifier. Hsu *et al.* [24] used the DL to depict how the local and global information can be used together for decision making in the puzzle-sets. Zuo *et al.* [25] used DL to detect human skin colour. They read the basic CNN-based architectures are insufficient for determining the connection between pixels and their neighbours. So, they introduced recurrent neural networks (RNNs) layers and integrated them into the fully CNNs (FCNs) to develop accurate results. Tajbakhsh *et al.* [26] developed a technique for the medical field application like cardiology, radiology, and gastroenterology using CNN. The pre-trained networks are retrained for fine-tuning to obtain the exact results. DNN proved a boon to the medical field because of its quality for learning the features automatically. Saha and Chakraborty [27] used DL to classify the cell membranes and nuclei that further helps in the evaluation of breast cancer. Ouyang *et al.* [28] used the deep networks to find the copy-moved tampered region of the image. The proposed algorithm uses the pre-existing model that was trained with the large database as ImageNet, and the net structure is set accordingly with the help of small training samples.

From the survey, we find out that initially, traditional machine learning methods are used for forgery detection, and now DL

methods are used to detect forgeries. The main reason to use the DL method is automatic forgery detection with less computation complexity of extensive data on the Internet. Nowadays, everyone owns a mobile phone with the Internet; therefore, they upload pictures and other media on the web. Most of the pictures are modified on the Internet, and the authenticity of these images is essential.

3 Proposed algorithm

In this paper, CMF and splicing, forgery are detected using a hybrid copy move and splicing image forgery detection using a machine learning-based colour illumination approach. To find forged and not forged images, a DCNN is used. The massive dataset is used for training in the DCNN model. The training is performed using supervised labelled images from the database. Our following task is to determine the mesh layer, the number of filters, and their size. In the subsequent step, the training options are defined, such as momentum, initial learn rate, learn rate schedule is piecewise, L2 regularisation, max epochs, and mini-batch size. Finally, the network is ready to train. The DNN is validated and tested with minimum image batch size. In the final step, the colour illumination-based ML method is used to detect passive forgeries from public datasets. It has two utmost significant steps, i.e. classification of images and detection of the doctored region. The performance is calculated using precision (P), recall (R), and $F1$.

$$P = \frac{\omega \cap \hat{\omega}}{|\hat{\omega}|} \quad (1)$$

where ω is the obtained forgery region, and $\hat{\omega}$ is the actual forgery region.

$$R = \frac{\omega \cap \hat{\omega}}{|\omega|} \quad (2)$$

where ω is the obtained forgery region, and $\hat{\omega}$ is the actual forgery region.

$$F1 = 2 \times \frac{R \times P}{R + P} \quad (3)$$

where $F1$ is the function of P and R , P is the precision, and R is the recall.

3.1 Proposed algorithm block diagram

This block diagram consists of hybrid DL and ML approaches to classify and detect forgeries. DL is used to classify forgeries, and ML-based colour illumination algorithm localises forgeries. DCNN learns about the forged images from the in-depth, gaining knowledge of the labelled image. DCNNs have newly accomplished extraordinary results in a diversity of computer vision (CV)-based areas. CNNs require an extensive labelled training dataset consisting of either images or feature extracted files from where the whole network is going to learn about the small details of the object in the given input.

The transfer learning approach is used to train a DCNN for multiple datasets. In each dataset, 80% of images are placed in the train folder, and 20% of images are placed in the test folder. Fig. 3 represents the proposed block diagram of the hybrid copy move and splicing image forgery detection using an ML-based colour illumination approach.

Transfer learning is used to train a DCNN model for multiple datasets. This model predicts two categories of copied images and not copied images. DNN (XONet) is trained on CASIAv1.0, CASIAv2.0, DVMM, and BSDS300 datasets with a base batch size of 20. All of these datasets consist of CMF and SF. CASIA-1 points on splicing forgery. This dataset contains 800 real shading pictures, and 921 spliced with a span of 384×256 pixels spared in JPEG format. The legitimate pictures are additionally divided into different classes as per the scene or questions within them like creature, character, surface, plant.

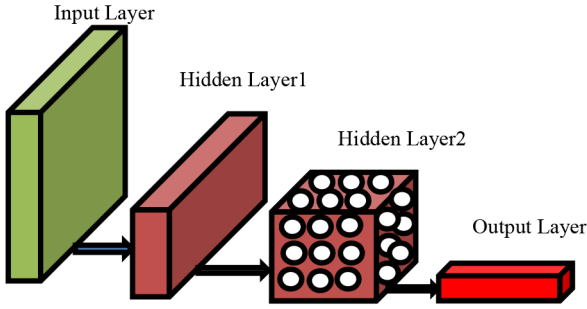


Fig. 4 Convolution-based DNN

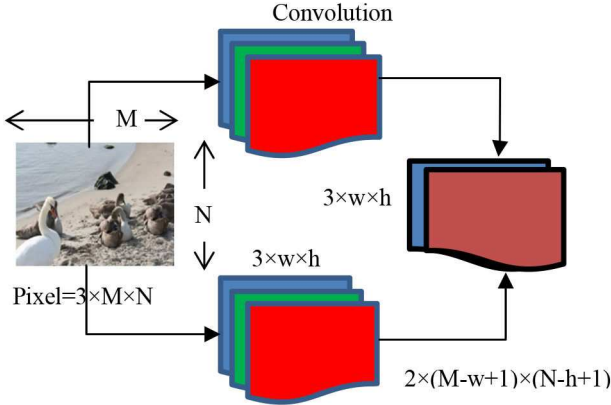


Fig. 5 Convolution with the kernel

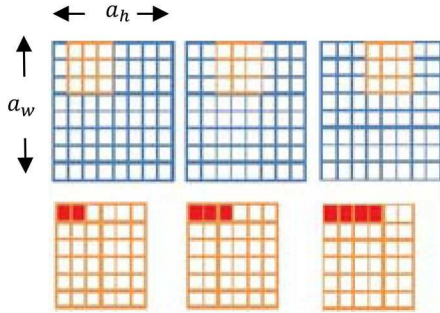


Fig. 6 Convolution between colour image and kernel with a stride of one

In the accompanying investigations, the dataset [29] is utilised to find out the CMF. This dataset has 48 high resolutions uncompressed PNG real nature images. The average size of the image is 1500×1500 . Images consist of the dataset are from the classes of living, nature, human-made, and blended. Different attacks such as scaling, rotation, JPEG compression and downsampled are applied to these images. The total number of CMF images is 1826 in this database. The DNN is the advanced version of ML, where the features are automatically extracted, and the information from them is learned at each increasing layer of the network. The DL method detects forgery automatically with less computation complexity for the massive data on the Internet. A general block diagram showing the CNN model is given in Fig. 3.

The dataset with millions of images are used to train the network sufficiently to its depth, but training a DCNN needs computationally heavy and large memory resource; otherwise, training would be tremendously time consuming. Furthermore, concepts like overfitting and convergence often complicate the training of the network. Hence, we need high-end GPUs to deal with this trouble. Thus, DL from scratch sometimes proves to be a time-consuming task, so some researchers see an option to use the already trained models for further processing.

Fig. 3 shows the step-by-step procedure to find out forgery. This block diagram is divided into two parts. The first part is used DCNN to predict image forgery, and the second part uses the ML-based colour illumination technique to detect forged patches. The proposed DCNN consists of 14 layers. These layers consist of one

input image layer, three convolutional layers, four ReLU layers, three max-pooling layers, two fully connected layers, and one softmax layer. The weights are then assigned to the first convolutional layer with 32 simple high-pass filters. The 32 filters are used in the first classification, and 16 filters are used in the second classification. The first convolutional layer consists of three colour channels, 32 output feature maps, with 5×5 filter size of weight matrices, then 94 weight matrices are required to initialise 32 fundamental filters. The training options are taken in the following manner.

The input layer is always considered as the first layer. This layer is used to define the size of the input images. Typically, an extensive dataset with millions of small size images is used to train the network. A small size image can be processed faster. The size of the image input is $116 \times 116 \times 3$. In the convolutional layer, 32 high-pass filters have been utilised for the initialisation of the weights. Each layer of the CNN is further connected to another neuron in the succeeding layer.

3.2 Convolution

The CNN has an input layer, multiple hidden layers, and the output layer. The input layer has several neurons connected to hidden layers. The input greyscale image consists of only one channel, and its depth is one. The input colour image consists of three colour channels, and its thickness is three. Let the colour image with three colour channels. M , N are the number of columns and rows, respectively. Then the total number of pixels present in this image is $3 \times M \times N$. This image has three colour channels so that we will have a three-dimensional matrix. Input image and kernel should have the same colour channel to get a three-dimensional matrix. If we perform special convolution of a colour image with the kernel $3 \times w \times h$, then we get one output. Fig. 4 shows the simple DNN convolution operation.

Similarly, if we perform the unique convolution of a colour image with the kernel $3 \times w \times h$, then we get another output. The size of the output channel is $2 \times (M-w+1) \times (N-h+1)$, where 2 represents the colour channel; M represents the width of the image; w represents the kernel; N represents the height of the image, h represents the kernel. Fig. 5 shows the convolution operation with the kernel.

3.3 Stride

The size of the image is reduced using stride operation. Fig. 6 shows the colour image having a pixel value x_k at the location, k is convolved with weight w_k at location k of the kernel. The equation gives the output of the function:

$$\text{Output} = b + \sum_k w_k \times x_k \quad (4)$$

where b is the bias of the neuron, x_k is the colour image pixel value, w_k is the weight of the kernel. This operation gives output for one pixel.

Similarly, we slide the kernel throughout the image, and we achieve all pixel values. The convolution of the 8×8 pixels with 3×3 kernel and stride one gives the output of 6×6 matrices. The calculation for the output width and height matrix is given by (2) and (3), respectively,

$$a_w = \frac{M - w}{S_w} + 1 \quad (5)$$

where M , w are the width and kernel, respectively, S_w is the stride value and a_w is the output matrix.

$$a_h = \frac{N - h}{S_h} + 1 \quad (6)$$

where N , h are the height and kernel, respectively, S_h is the stride value and a_h is the output matrix.

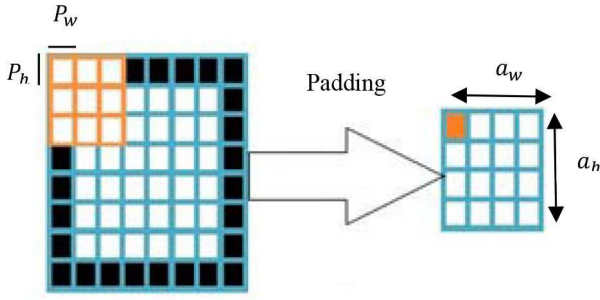


Fig. 7 Zero paddings to perform convolution between 7×7 image matrix and 3×3 kernel

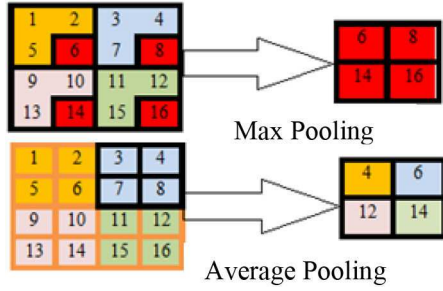


Fig. 8 Maximum pooling and average pooling operation



Fig. 9 Classification of not spliced (shown as Sp in red colour) and spliced image (shown as Sp in green colour)

3.4 Padding

The image matrix size changes during this operation, so the zero paddings are used to match the image matrix dimensions. Let us take an example to perform convolution between the 7×7 image matrix and 3×3 kernel. In this case, zero paddings are used to complete this operation as shown in Fig. 7.

$$o_w = \frac{M - w + 2p_w}{S_w} + 1 \quad (7)$$

where M , w are the width and kernel, respectively, S_w is the stride value, $2p_w$ is the padding width and o_w is the output matrix width.

$$o_h = \frac{N - h + 2p_h}{S_h} + 1 \quad (8)$$

where N , h are the width and kernel, respectively, S_h is the stride value, $2p_h$ is the padding height and o_h is the output matrix height.

3.5 Pooling

The size of the image is reduced using a pooling operation. In pooling, we look into a small neighbourhood and pool it out with one single value. This process is aggregating all values into one single value. Fig. 8 shows the max and average pooling operations.

3.6 Rectified linear unit (ReLU) transfer function

ReLU is used to convert positive values to 1 and negative values to 0. The derivative of any negative value is zero, and the derivative of any positive value is unity. Foremost, the dataset with properly labelled pictures is prepared. As we have experienced, recalling an image means recalling a matrix. This matrix consists of pixel values range from 0 to 255. In supervised learning, mapping of the input information to the output label is managed.

Further, we perform the convolution between the input image and kernel. The size of the kernel is defined on the type of objects in the image. The extracted features from the image are applied as an input to the next layer. Each layer of the DCNN is connected to the next layer.

The convolution process will make a considerable matrix. This size of the matrix will be challenging to process further. Pooling is used to cut down the size of the matrix. It reduces the execution difficulty of the model. The next step is performing probability conversion. High numeric values of images in the matrix are utilised to make the probability of occurrences of a particular picture element in the picture. The softmax function is utilised to assign a probability.

3.7 Image forgery classification

Copy move and SF classification are performed using DCNN, which classify image forgery in terms of forged and not forged.

After the classification process is over, we get a performance on the validation sets. The classification algorithm produces the output as correctly classified forged and not forged images, as shown in Fig. 9.

3.8 Illuminants for detecting forgeries

Input: Take forged image; **Output:** Colour of the illuminant.

STEP-1: Let $f(\text{pix}) = (f_{\text{red}}(\text{pix}), f_{\text{green}}(\text{pix}), f_{\text{blue}}(\text{pix}))^T$; RGB colour.

STEP-2: Perform colour illuminants, $f(\text{pix})$ is formed by $F(\text{pix}) = \int_{\Omega} e(\lambda, \text{pix}) s(\lambda, \text{pix}) c(\lambda) d\lambda$ for detecting forgeries.

STEP-3: Put forged pix values in [30] model to get the colour illuminant.

Let $f(\text{pix}) = (f_{\text{red}}(\text{pix}), f_{\text{green}}(\text{pix}), f_{\text{blue}}(\text{pix}))^T$, where $f(\text{pix})$ is the forged pix. In [30] they have considered all conditions idle while capturing images

$$F(\text{pix}) = \int_{\Omega} e(\lambda, \text{pix}) s(\lambda, \text{pix}) c(\lambda) d\lambda \quad (9)$$

where Ω represents measurable illumination, illuminant spectrum is represented by $e(\lambda, \text{pix})$, object surface reflectance is represented by $s(\lambda, \text{pix})$, λ is the illumination wavelength, and camera colour sensitivities is represented by $c(\lambda)$.

$$ke^{n, p, \sigma} = \left(\int \left| \frac{\partial^n f^{\sigma}(\text{pix})}{\partial x^n} \right|^p d_{\text{pix}} \right)^{1/p} \quad (10)$$

where $|\cdot|$ is the absolute value, x , k are the pixel coordinate and scaling factor, respectively, a differential operator is represented by ∂ , and observed intensities are represented by $f^{\sigma}(\text{pixel})$.

3.9 Adaptive over segmentation

Input: Take colour of the illuminant image; **Output:** Adaptive over segmented image.

$$\text{STEP} - 1: ELF = \sum |CA4|, \quad (11)$$

$$\text{STEP} - 2: EHF = \sum (\sum |CDi| + \sum |CHI| + \sum |CVi|), \quad (12)$$

$$\text{STEP} - 3: PLF = \frac{ELF}{(ELF + EHF)} \times 100 \%, \quad (13)$$

$$STEP - 4: S = \sqrt{(0.02 \times Q \times R)}; PLF > 50\%, \quad (14)$$

$$STEP - 5: S = \sqrt{(0.01 \times Q \times R)}; PLF \leq 50\%, \quad (15)$$

$$STEP - 6: \text{Initial size } S = Q \times R \text{ of the host image.} \quad (16)$$

3.10 Simple linear iterative clustering (SLICO)

Input: Adaptive over segmented image; **Output:** Simple linear iterative clustering (SLICO) image.

STEP-1: Select the maximum value of colour distance.

STEP-2: Give the maximum colour distance for a cluster.

STEP-3: Calculate the centroid and store in the seed values.

STEP-4: Perform the colour conversion; find seeds; compute superpixels.

STEP-5: Enforce connectivity; assign output labels; assign several labels/seeds.

3.11 Scale-invariant feature transform (SIFT)

Input: SLICO image; **Output:** SIFT image.

STEP-1: Load grayscale image I.

STEP-2: Create four $\times K$ matrices to store one SIFT frame per column.

STEP-3: Algorithm returns the difference of Gaussians scale-spaces.

STEP-4: Clear image boundaries and descriptor intersects boundary.

STEP-5: Threshold must be higher than zero or 0.01; features below this threshold are ignored.

STEP-6: Edge threshold must be greater than or equal to zero; features above this threshold are ignored.

3.12 Block feature matching

Input: SIFT image; **Output:** Block feature matching.

STEP-1: Matching between patches. Set the threshold distance between detected SIFT pixels is 0.15.

STEP-2: Create two patches A, B , and find out key points x, y for A , and B , respectively.

STEP-3: Perform key point threshold calculation and give the decision for correct key points.

STEP-4: Perform colour grow operation using the threshold colour intensity value of red, green, and blue components.

STEP-5: Perform morphological operation to get correct forgery regions.

3.13 Performance calculation

Performance accuracy is calculated at two levels to evaluate the algorithm. Firstly, it is calculated on the image forgery classification stage. Secondly, it is calculated on the image forgery localisation stage. After training with multiple datasets of CMF and splicing images, our algorithm (see Fig. 10) gives prediction accuracy. Tables 1 and 2 show the training of DCNN using transfer learning approach for multiple datasets. The performance accuracy is calculated on the CASIA v1.0 validation set, and the test set is 0.9807 and 0.9878, respectively. The performance accuracy is calculated on the CASIA v2.0 validation set, and the test set is 0.9767 and 0.9805, respectively.

In the image forgery localisation stage, our algorithm gives a performance, accuracy using (1–3) in terms of precision = 97%; recall = 100% and $F1 = 99\%$ for CoMoFoD dataset.

4 Proposed algorithm results

In this paper, the number of experiments is performed on different data sets. These experiments are performed on GPU enabled machine, which has a specification as Intel i7 processor; Z170X

Step-1: Use image Datastore for loading the two image categories

Step-2: Classify with a trained DCNN model named as XONet into labels and error values

Step-3: Create a confusion matrix of labels from the test dataset into 2×2 matrix value

Step-4: Calculate the value of the confusion matrix

Step-5: Calculate the mean value of the diagonal in the confusion matrix

Fig. 10 Algorithm 1

Table 1 DCNN training on CASIA v1.0 using transfer learning

Epoch	Mini-batch loss	Mini-batch accuracy, %	Time elapsed, s	Iteration
1	0.6931	50.00	2.27	1
50	0.4314	85.00	291.18	550
100	0.1377	100.00	582.36	1100
150	0.1184	100.00	878.15	1650
200	0.1166	100.00	1172.63	2200

Table 2 DCNN training on CASIA v2.0 using transfer learning

Epoch	Mini-batch loss	Mini-batch accuracy, %	Time elapsed, seconds	Iteration
1	0.6932	25.00	1.86	1
50	0.1409	90.00	11,612.89	22,200
100	0.0050	100.00	23,513.76	44,850
150	0.0101	100.00	35,397.85	67,500
200	0.0027	100.00	47,384.30	90,150

Gaming G1 motherboard with 32 GB DDR4 RAM; PCIe Gen 3 $\times$ 4 SLI slots with the speed of the 32 GB / s; PCIe Gen2 $\times$ 2 slots with 10 GB/s data transfer speed; NVIDIA SLI four-way, three-way and two-way bridge; NVIDIA GEFORCE GTX 1070 $\times$ 4 GPU with 8GB $\times$ 4 = 32 GB GPU RAM.

Image forgery classification and localisation using the proposed approach on different datasets [3, 4, 29] are carried out in MatLab 18 software. In the database [29], 48 images of plain CMF are available along with ground truth images. These images are evaluated, tested, and compared with different parameters to the existing technique [30–32]. In the dataset [4], spliced images are present. CASIA 1.0 and 2.0 versions have Authentic and Forged image folders. In the CASIA 1.0 version's authentic folder, 800 images are present, and in CASIA 2.0 version authentic folder 7491 images are available. In the CASIA 1.0 version's spliced folder, 800 images are present, and in CASIA 2.0 version spliced folder, 7491 images are available. CMF and splicing are two types of forgeries, which are available in the dataset BSDS. In this dataset, we have taken 100 images for test and 200 images to train the network.

The result is shown in Fig. 11, row1, for the CASIA v1.0 database. In the CASIA v1.0 authentic folders contain 800 images, and in a spliced folder, 921 images are present. These images are divided into two folders, such as train and test. Training images are of 80%, and test images are of 20%. All SF images are shown in the first row (top). The image region analyser application creates forgery detected output shown in the first row (bottom). Using the algorithm performance accuracy is calculated on the CASIA v1.0 validation set, and the test set is 98 and 99%, respectively.

The result is shown in Fig. 11, row2, for the CASIA v2.0 database. In the dataset, CASIA v2.0 authentic folder contains 7491 images, and in a spliced folder, 5123 images are available. All SF images are shown in the second row (top). Image region analyser application creates forgery detected output shown in the second row (bottom). Using the algorithm performance accuracy is calculated on the CASIA v2.0 validation set, and the test set is 97 and 98%, respectively.

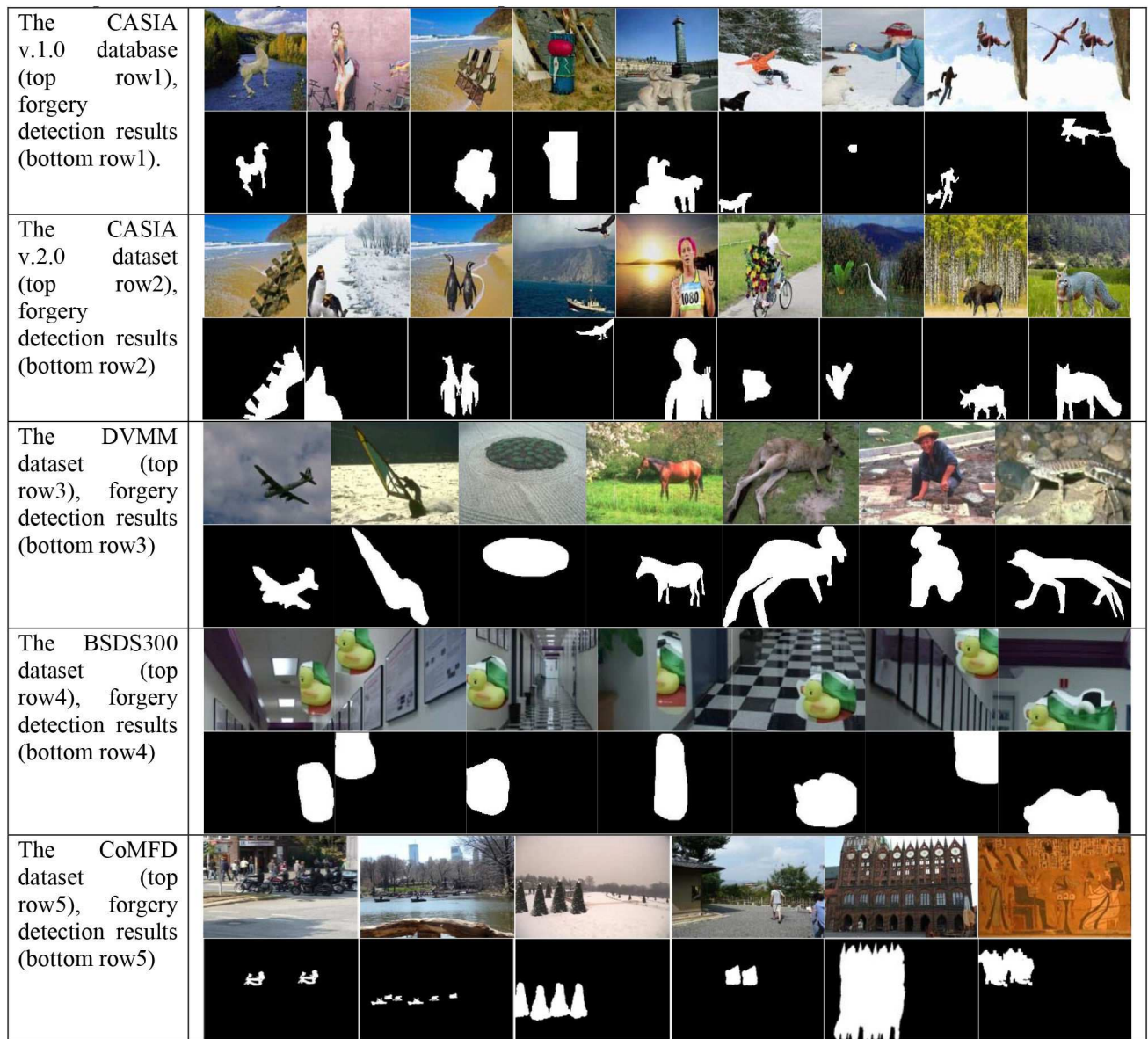


Fig. 11 CASIA-1 (row 1), CASIA-2 (row 2), DVMM (row 3), BSDS300 (row 4), CoMFD (row 5) datasets of CMF and SF detection results

Table 3 CMF and SF accuracy comparison

Authors	CASIA1 Acc., %	CASIA2 Acc., %	DVMM Acc., %	BSDS Acc., %	CMF Acc., %
proposed	99	98	97	98	99
Zhao <i>et al.</i> [33]	—	—	94	—	—
He <i>et al.</i> [34]	—	90	94	—	—
Muhammad <i>et al.</i> [35]	95	97	—	—	—
Rao and Ni [23]	98	98	96	—	—

The result is shown in Fig. 11, row3, for the DVMM database. Splicing and CMFs are two types of forgeries, which are available in the DVMM dataset. The dataset contains 100 images for testing and 200 images for training the network. In the third row (top), all spliced images are shown. In the third row (bottom) image region analyser application creates forgery detected output. The DVMM dataset forgery detection accuracy is 97%.

The result is shown in Fig. 11, row4, for the BSDS300 database. In this dataset, 200 images are available for training, and 100 images are present for testing. In the first row, all spliced images are shown. In the fourth row (top) machine learning-based colour illumination forgery detected, regions are shown. In the fourth row (bottom) image region analyser application creates, forgery detected output. The BSDS300 dataset forgery detection accuracy is 98%.

The result is shown in Fig. 11, row5 for CoMFD dataset. This dataset contains 48 images of plain copy-move forged images. This dataset contains natural, architects, animal, art, plant, and text images. In the fifth row (top), all copy-move forged images are shown. In the fifth row (bottom), morphological operation shows forgery detected output. The proposed algorithm is tested on image-level using the achieved performance accuracy values, i.e. precision (P) = 98%, recall (R) = 100% and $F1 = 99\%$.

Table 3 shows the accuracy result comparison on CASIA 1, CASIA 2, DVMM, BSDS300, and CMFD datasets. From the comparison, we can say that our proposed method performs better in all datasets. Fig. 12 shows the bar graph between proposed and other methods. From Fig. 12 blue colour represents the proposed method; orange colour represents Yuan Rao; yellow colour represents Muhammad; violet colour represents He; green colour

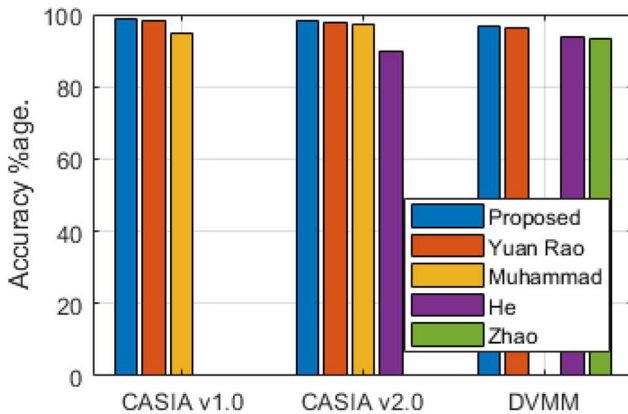


Fig. 12 Bar graph for CMF and SF accuracy comparison

represents Zhao. The forgery detection accuracy is better as compared to all other states of the methods.

5 Conclusion

A hybrid DL and ML approach for passive image forensic is proposed in this paper. The DCNN classifies images into forged and not forged categories. The pre-trained DCNN model use, transfer learning approach to learn more image patterns. With the help of DCNN, we extract features from test images and classify image categories. The machine learning using a colour illumination algorithm localises CMF and splicing forgery. From the experimental results shown in Fig. 11, the localisation of the forged areas is detected correctly.

The performance accuracy is calculated on the CASIA v1.0 validation set, and the test set is 98 and 99%, respectively. The performance accuracy is calculated on the CASIA v2.0 validation set, and the test set is 98 and 98%, respectively. The DVMM dataset forgery detection accuracy is 97%. The BSDS300 dataset forgery detection accuracy is 98%. The proposed algorithm is tested on an image level on CMFD dataset and achieved performance accuracy, i.e. precision (P) = 98%, recall (R) = 100% and $F1$ = 99%.

6 References

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